ENV S 193DS Final

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[Github Repo] (https://github.com/nvanderpoel-cmd/ENVS-193DS_spring-2026_final)

Set up

```
# reading in packages
library(here)
library(tidyverse)
library(janitor)
library(dplyr)
library(MuMIn)
library(DHARMa)
library(ggeffects)
library(ggplot2)
library(readxl)

# reading in data
nests <- read.csv(here("data/nest_data_final.csv"))
my_data <- read_xlsx(here("data/my_data.xlsx"))
```

Problem 1. Research Writing

a. Transparent statistical methods

b. Suggestions for rewriting

c. Further analysis

Problem 2. Data analysis

a. Clean and wrangle

```
# new object nests_clean
nests_clean <- nests |>
  select(height_cm, case_control, ant_species) |>
  filter(ant_species %in% c(
                      "AB",
                    "RRB",
                    "BBR")) |>
  mutate(ant_species = as_factor(ant_species),
         ant_species = fct_relevel(ant_species,
                                   "AB",
                                   "RRB",
                                   "BBR")) |>
  mutate(species_name = case_match(
    ant_species,
    "AB" ~ "Crematogaster sjostedti",
    "RRB" ~ "Crematogaster mimosae",
    "BBR" ~ "Crematogaster nigriceps"
  ))
```

```
Warning: There was 1 warning in `mutate()`.
i In argument: `species_name = case_match(...)`.
Caused by warning:
! `case_match()` was deprecated in dplyr 1.2.0.
i Please use `recode_values()` instead.
```

```
slice_sample(nests_clean, n = 10)
```

```
   height_cm case_control ant_species            species_name
1      181.0            1         RRB   Crematogaster mimosae
2      121.5            0         RRB   Crematogaster mimosae
3      128.0            0         BBR Crematogaster nigriceps
4      165.0            0         RRB   Crematogaster mimosae
5      168.5            0         RRB   Crematogaster mimosae
6      163.5            0         RRB   Crematogaster mimosae
7      110.5            0         RRB   Crematogaster mimosae
8      310.0            0         RRB   Crematogaster mimosae
9      167.0            0         BBR Crematogaster nigriceps
10     120.0            0         RRB   Crematogaster mimosae
```

b. Response variable

For the response variable, case_control, the 1's signify there is a nest, of any bird species, present, while a 0 signifies there is no nest present.

c. Purpose of study

Tree height: Tree height is a continuous variable. As tree height increases, the probability of a nest occurrence would increase because of increased distance between nests and browsing mammals.

Acacia ant species: Acacia ant species is a categorical variable.

d. Table of models

Model number	Tree Height	Ant Species	Interaction
1	X	X	X
2	X	X	

e. Run the models

```r
# model 1: Model with interaction term

model1 <- glm(case_control ~ height_cm * ant_species,
              data = nests_clean,
              family = "binomial"
```

```
              )

# model 2: Model without interaction term

model2 <- glm(case_control ~ height_cm + ant_species,
              data = nests_clean,
              family = "binomial"
              )
```

f. Select the best model

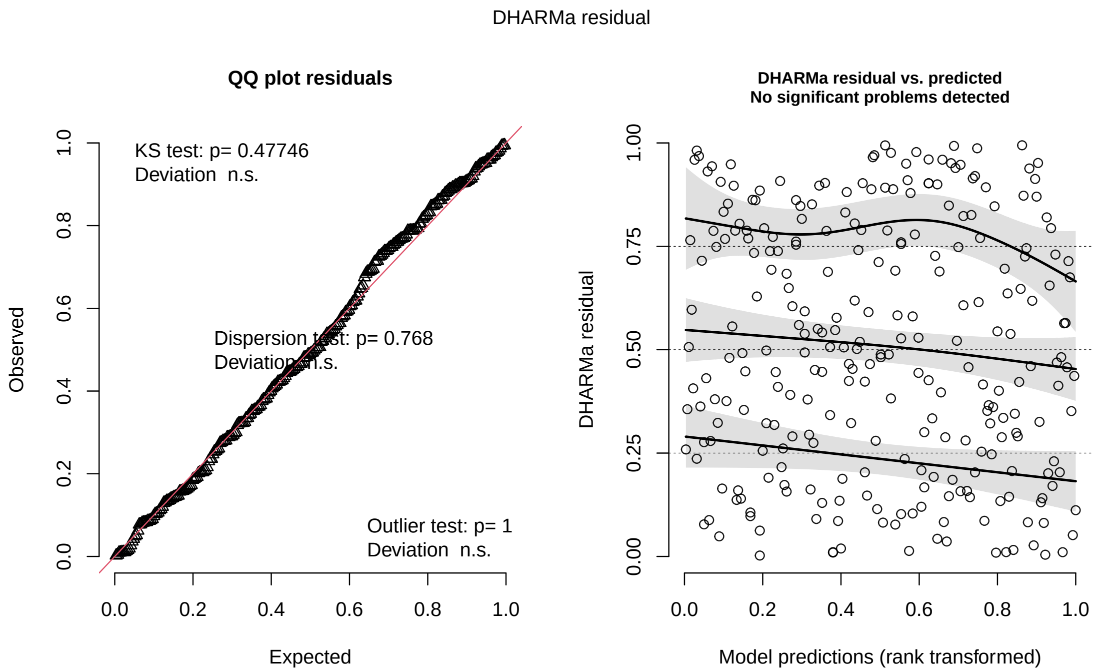
```
AICc(
model1,
model2
) |>
arrange(AICc)
```

```
       df     AICc
model1  6 238.1236
model2  4 258.8940
```

The best model as determined by Akaike's Information Criterion (AIC) was model 1, with interaction.

g. Check the diagnostics

```
plot(
  simulateResiduals(model1)
)
```



h. Visualize the model predictions

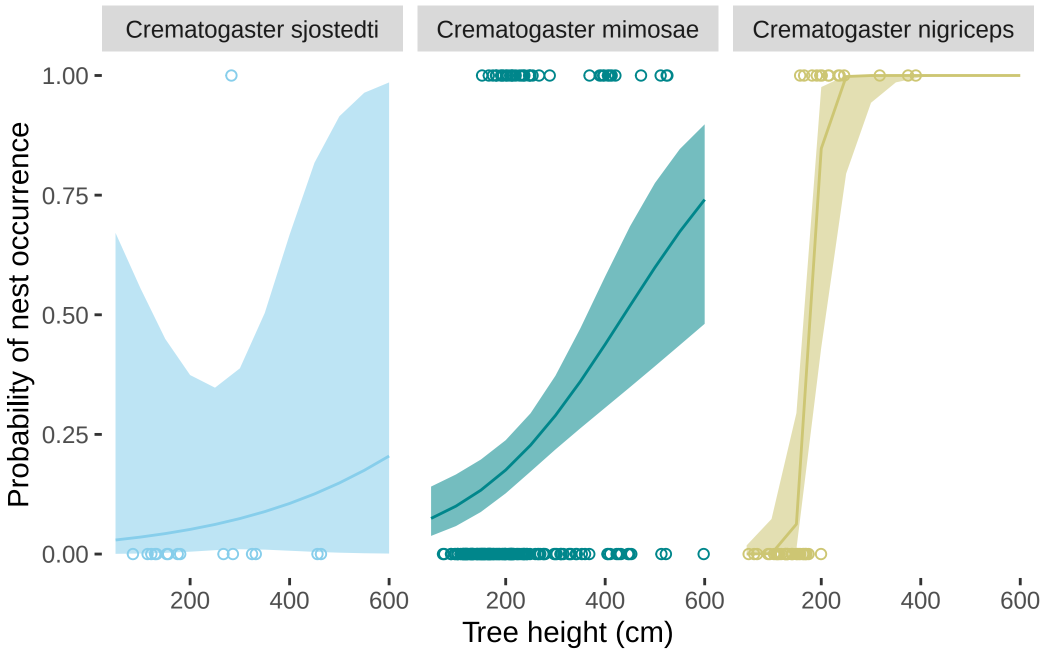
```
# creating new object mod_preds
mod_preds <- ggpredict(model1,
                       #adding predictor variables
                       terms = c("height_cm",
                                 "ant_species")) |>
  #renaming group column ant_species
  dplyr::rename(ant_species = group)
```

```
Data were 'prettified'. Consider using `terms="height_cm [all]"` to get
  smooth plots.
```

```
#ggplot base
# x = height, y = nest occurrence, categorized by ant species
ggplot(nests_clean,
       aes( x = height_cm,
            y = case_control)) +
  # adding points representing observations
```

```r
  geom_point(aes(color = ant_species),
             shape = 21) +
  # adding confidence level ribbon based on mod_preds
  geom_ribbon(data = mod_preds,
              aes(x = x,
                  y = predicted,
                  ymin = conf.low,
                  ymax = conf.high,
                  fill = ant_species,
              alpha = 0.1)) +
  # adding prediction line
  geom_line(data = mod_preds,
            aes( x = x,
                 y = predicted,
                 color = ant_species)) +
  # creating different panels
  # labelling by full name instead of abbreviations
  facet_wrap(~ant_species,
             labeller = labeller(
               ant_species = c(AB = "Crematogaster sjostedti",
                               RRB = "Crematogaster mimosae",
                               BBR = "Crematogaster nigriceps"))) +
  #custom colors
  scale_color_manual(values = c(
    "AB" = "skyblue",
    "RRB" = "turquoise4",
    "BBR" = "khaki3"
  )) +
  # matching ribbon colors
  scale_fill_manual(values = c(
  "AB" = "skyblue",
  "RRB" = "turquoise4",
  "BBR" = "khaki3"
)) +
  # adding x range
  # adding line breaks
  scale_x_continuous(limits = c(50, 600),
                     breaks = c(200, 400, 600)) +
  # getting rid of legend
  theme(legend.position = "none",
        panel.background = element_blank(), # removing grey background
        panel.grid.major = element_blank() # removing grid lines
```

```
        ) +
# labelling axes
  labs(x = "Tree height (cm)",
      y = "Probability of nest occurrence")
```



i. Write a caption for your figure

Figure 1: As height increases, the probability of bird nest occurrence increases with each acacia ant species. Data from nest_data_final.csv (Lujan E.et al, Zenodo, 2023). Lines represent predictions of probability of nest occurrence (lbs) given acacia species; Crematogaster sjostedti (AB), Crematogaster mimosae (RRB), Crematogaster nigriceps (BBR), (total n = 270) and tree height in cm. Meanwhile, points represent actual observations of if a nest occurred or not(0 = nest not present, 1 = nest present). The probability of nest occurrence increases as tree height increases. Nevertheless, the probability increases at different rates with each ant species, where Crematogaster sjostedti has the most gradual increase in probability followed by Crematogaster mimosae and Crematogaster nigriceps. Ribbons represent a 95% confidence interval. Colors represent acacia ant species (skyblue: Crematogaster sjostedti, teal: Crematogaster mimosae, khaki: Crematogaster nigriceps).

j. Calculate model predictions

```
ggpredict(model1,
          terms = c("height_cm [600]",
                    "ant_species"))
```

```
# Predicted probabilities of case_control

ant_species: AB

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.20 | 0.00, 0.99

ant_species: RRB

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.74 | 0.48, 0.90

ant_species: BBR

height_cm | Predicted |     95% CI
----------------------------------
      600 |      1.00 | 1.00, 1.00
```

k. Interpret your results

Acacia ant species and tree height (cm) interact to predict the probability of nest occurrence. As tree height increases, probability of nest occurrence increases.

Furthermore, at a tree height of 600 cm, the probability of nest occurrence is 0.20 for Crematogaster sjostedti (95% CI: [0.00, 0.99]), 0.74 for Crematogaster mimosae (95% CI: [0.48, 0.90]), and 1.00 for Crematogaster nigriceps (95% CI: [1.00, 1.00]). Indicating higher probability of nest occurrence for Crematogaster nigriceps at the highest tree height measured, 600 cm.

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

There are many differences but a large difference is the amount of observations and data points, also of course they are displayed very differently. The affective visualization is a lot more artistic and its hard to tell it is data at first. Both graphs converge however in showing the low amount of caffeine consumed the more sleep in hours was had. Nevertheless, the patterns differ somewhat between the two. The graph from homework 02 did not have any distinct patterns and looked relatively random although there were a couple points that showed a pattern of negative correlation starting to emerge. The affective visualization had a much more prominent decrease and much more data to go off of. From week 9, I received mostly suggestions about coloring and incorporated these through watercolor for my final product.

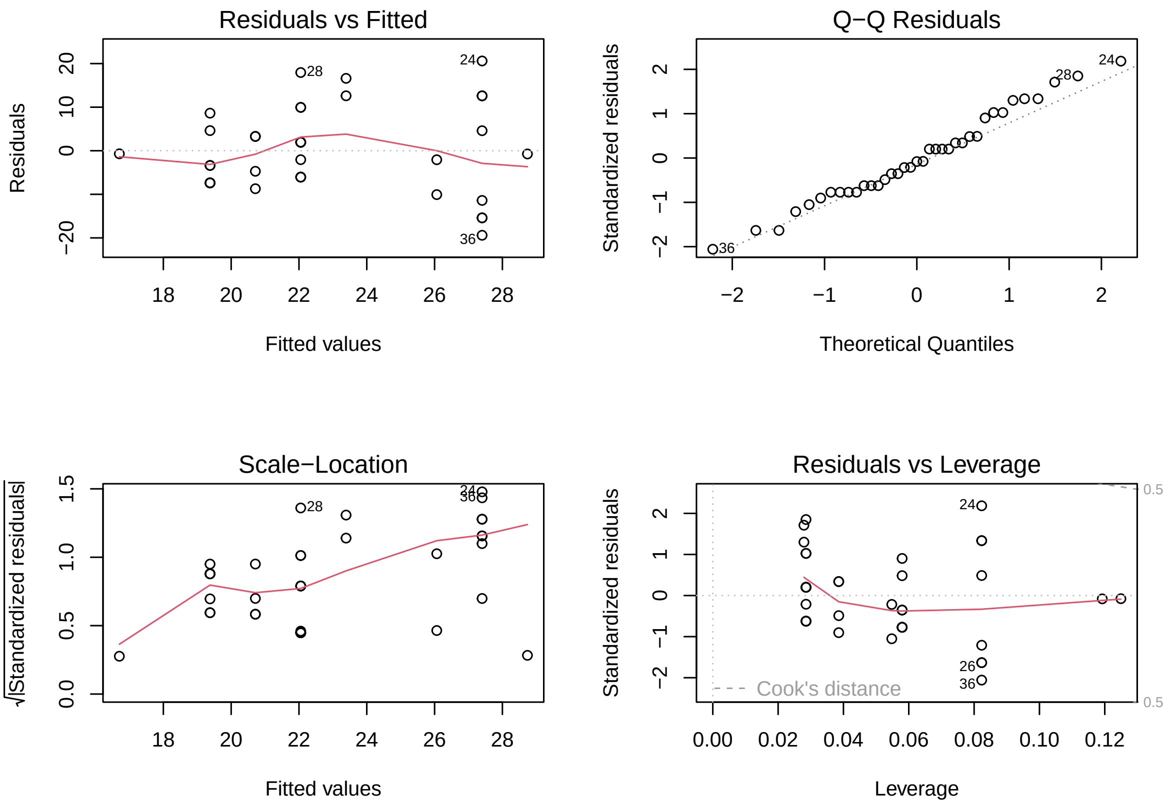
b. Designing an analysis

A Pearson's correlation could be applied to answer the research question: How does the amount of sleep in hours influence the amount of coffee consumed in ounces per day? Here, a Pearson's correlation is appropriate because it uses parametric tests measuring the mean, there are no significant outliers in the data, and the research question asks for the strength of correlation or how strongly the predictor influences the response variable.

c. Check any assumptions and run your analysis

Checking assumptions

```
my_data_clean <- clean_names(my_data)

# creating new object
# linear regression model of my data
# (response ~ predictor)
my_data_lm <- lm( coffee_oz ~ amount_of_sleep,
    data = my_data_clean
    )
#displaying linear model
# putting side by side
par(mfrow = c(2,2))
# plotting linear model
plot(my_data_lm)
```



Running Pearson's Correlation

```r
# running pearsons coefficient
# data$predictor, data$response
cor.test(my_data_clean$amount_of_sleep, my_data_clean$coffee_oz,
         method = "pearson")
```

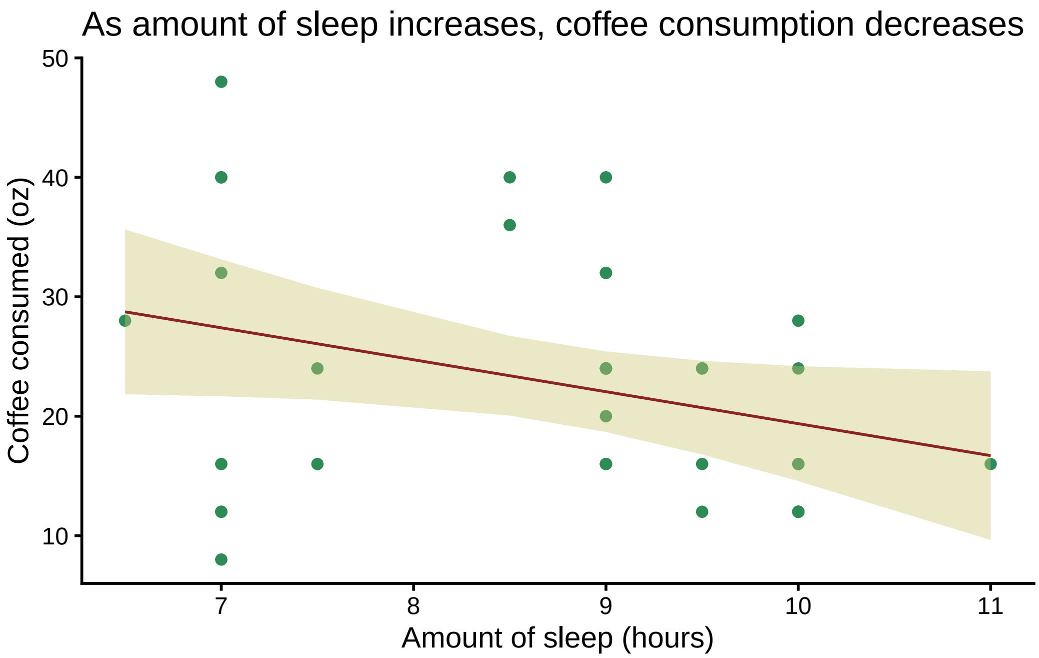
```
	Pearson's product-moment correlation

data:  my_data_clean$amount_of_sleep and my_data_clean$coffee_oz
t = -1.9835, df = 35, p-value = 0.0552
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.581958286  0.006843215
```

```
sample estimates:
       cor
-0.3178806
```

d. Create a visualization

```
# creating new object, my_data_preds
my_data_preds<- ggpredict(my_data_lm,
                          terms = "amount_of_sleep")
# base layer ggplot
# x = amount of sleep
# y = coffee consumed
ggplot(my_data_clean,
       aes(x = amount_of_sleep,
           y = coffee_oz)) +
  # adding my data in points
  geom_point(color = "seagreen") +
  # adding confidence interval ribbon
  geom_ribbon(data = my_data_preds,
              aes(x = x,
                  y = predicted,
                  ymin = conf.low,
                  ymax = conf.high),
              # ribbon color
              fill = "khaki3",
              alpha = 0.4) +
  # adding fitted linear model
  geom_line(data = my_data_preds,
            aes( x = x,
                 y = predicted),
            # line color
            color = "brown4") +
  # labelling axes and title
  labs(
    x = "Amount of sleep (hours)",
    y = "Coffee consumed (oz)",
    title = "As amount of sleep increases, coffee consumption decreases"
  ) +
  # changing theme from default
  theme_classic() +
```

```r
  # getting rid of legend
  theme(legend.position = "none")
```



e. Write a caption

Figure 2: As amount of sleep increases, coffee consumption decreases. Data my_data.xlsx (van der Poel, 2026). Line represents fitted line or linear model, demonstrating relationship between amount of sleep (hours) and coffee consumed (oz) (total n = 37). Meanwhile, points represent actual observations of coffee consumed on a given day given a certain amount of sleep. The amount of coffee consumed decreases as amount of sleep increases. Ribbon represents a 95% confidence interval. Colors represent different graph elements (brown: fitted line, seagreen: observations, khaki: confidence interval ribbon).

f. Write about your results

I found a weak negative relationship between amount of sleep and coffee consumed (Pearson's r = 0.32, t(35) = -2.0, p = 0.055, $\alpha = 0.05$).

g. Sharing your affective visualization